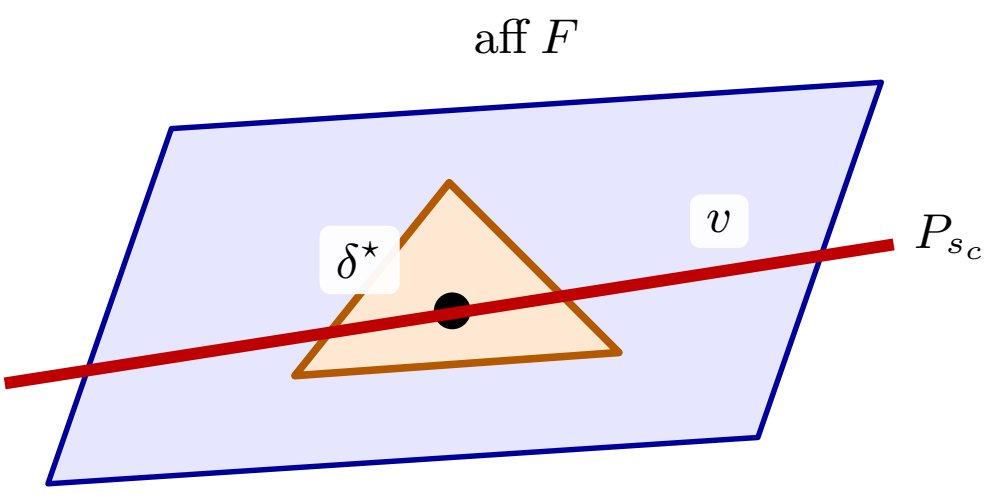
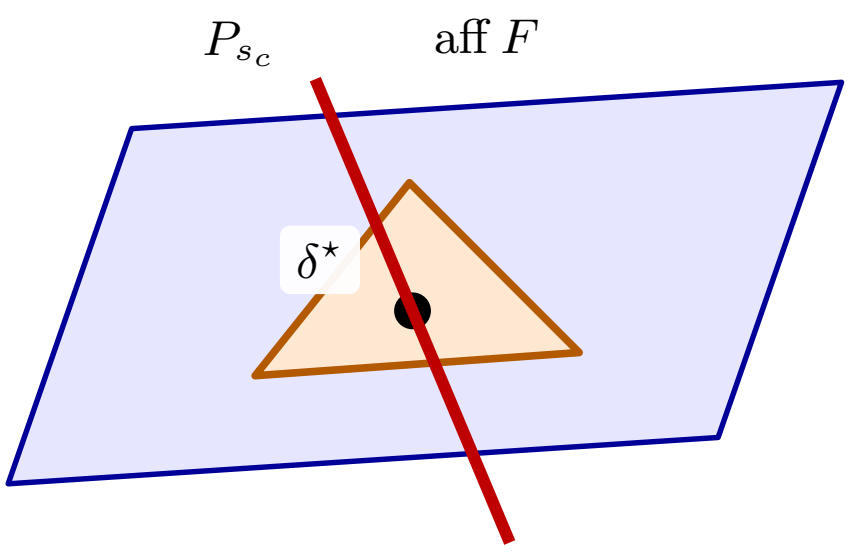


Case A: shared direction

$\dim(P_{s_c} \sqcap \mathrm{dir}) \geq 1$



Case B: transverse

$\dim(P_{s_c} \sqcap \mathrm{dir}) = 0$